EVALUATING LIMITS GRAPHICALLY

LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- Explain how graphical representations provide an intuitive approach to grasp the concept of limits.
- Differentiate between continuous and discontinuous functions and identify the graphical cues that indicate continuity or discontinuity.
- Apply graphical techniques to evaluate one-sided limits and recognize their graphical representation.
- Identify various types of discontinuities (removable, jump, infinite) in functions and explain their graphical characteristics.

INTRODUCTION

Graphs provide an intuitive way to visualize the behavior of functions as they approach a specific point. This visual representation helps students develop an intuitive understanding of limits.

Graphs make it easier to spot discontinuities, such as holes, jumps, or infinite behaviors, which are critical in limit evaluation. Discontinuities are often visually apparent on graphs.

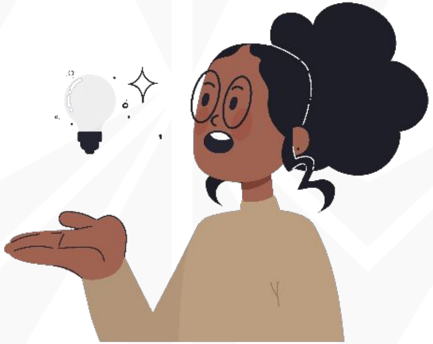
INTRODUCTION

Graphs also help us grasp the concept of one-sided limits by visually seeing how the function approaches a point from the left and right sides.

In various fields like physics, engineering, economics, and computer science, graphical understanding of limits is essential for modeling and solving real-world problems.

CONNECTION BETWEEN CONTINUITY AND LIMIT

Why do we talk about continuity, when we talk about graph of a limit?



WHAT IS CONTINUITY?

Continuity of a function at a point is defined using limits. A function $f(x)$ is continuous at a point c in its domain if the following conditions are met:

The function is defined at c, meaning f(c) is defined.

The limit of the function as x approaches c exists ($\lim_{x \to c} f(x)$ exists).

The **limit of the function** as x approaches c is equal to the **value of the function** at c ($\lim_{x \to c} f(x) = f(c)$).

LIMIT AND CONTINUITY

Limits play a crucial role in determining if a function is continuous at a specific point or not.

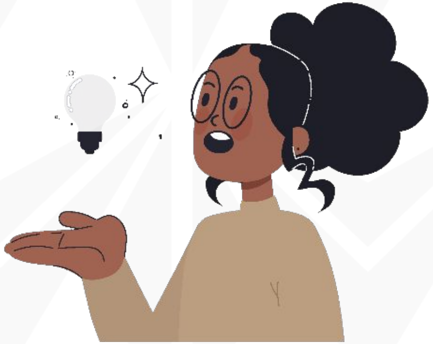
For a function to be continuous at a point c, the limit of the function as x approaches c must exist and be equal to the value of the function at c.

If the limit of the function at c does not exist, or if it exists but is not equal to $f(c)$, the function is **not continuous** at c.

CONTINUITY ON A GRAPH

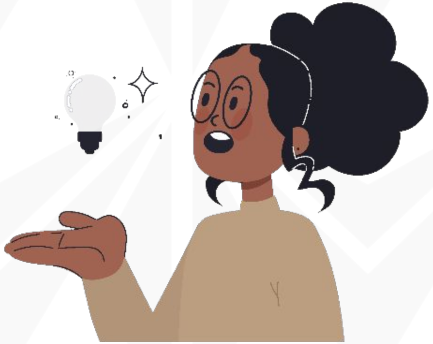
On a graph, continuity is visually represented by a smooth, unbroken curve. At a point of continuity, you can draw the graph without lifting your pen.

If there is a hole, jump, or any disruption (on that line you're trying to draw) in the graph at a particular point, it indicates a discontinuity, and the function is not continuous at that point.



QUESTION

What is discontinuity of a function?



TYPES OF DISCONTINUITY

Discontinuities can be identified and analysed using limits. Common types include:

1. Removable Discontinuity

2. Jump Discontinuity

3. Infinite Discontinuity

4. Essential Discontinuity

RECALL THAT

We often consider two types of one-sided limits: the limit from the left and the limit from the right.

In order for us to analyse discontinuity, we check how the function approaches a point from both sides.

If the approaches differ, it's a discontinuity.

HOW TO FIND THE VALUE OF A LIMIT FROM A GRAPH

In order for us to check the continuity or discontinuity of a function *f*, given a graph of *f(x)*,

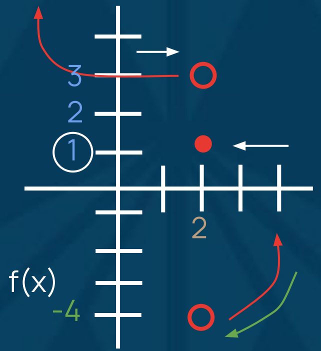
$$\lim_{x \to 2} f(x)$$

We will check from the left and right i.e.

$\lim_{x \to 2^-} f(x) = 3$ and $\lim_{x \to 2^+} f(x) = -4$

HOW TO FIND THE VALUE OF A LIMIT FROM A GRAPH

If $\lim_{x\to 3+} f(x) = \lim_{x\to 3-} f(x)$, the limit is said to **EXIST** but if $\lim_{x\to 3+} f(x) \neq \lim_{x\to 3-} f(x)$, we say the limit DOES NOT EXIST (DNE).



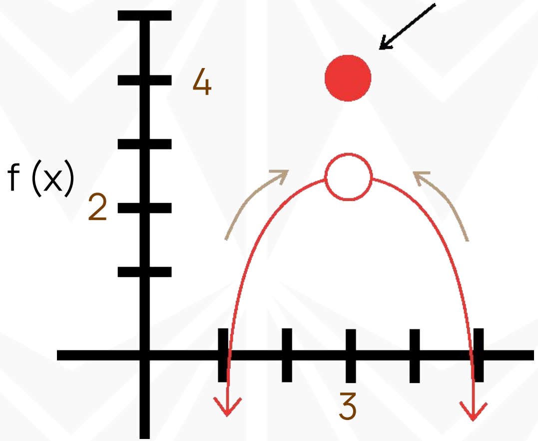
To find $f(x)$, we check the shaded dot towards the y-axis i.e. $f(2) = 1$

TASK 1

$\lim_{x \to 3^-} f(x) = ?$ $\lim_{x \to 3^+} f(x) = ?$

Then, $\lim_{x \to 3} f(x) = ?$

Also, what is $f(3) = ?$



GRAPH OF INFINITY

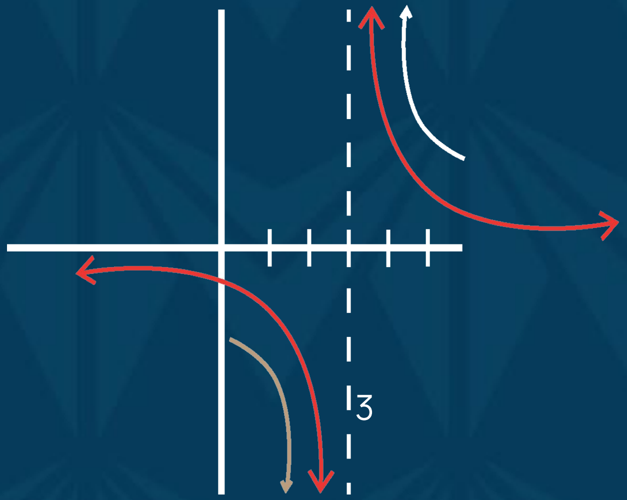
$$\lim_{x \to 3^+} f(x) = +\infty$$

$$\lim_{x \to 3^-} f(x) = -\infty$$

Therefore,

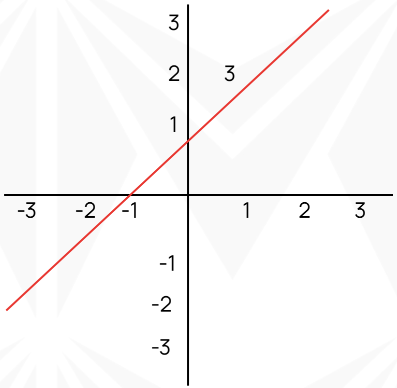
$$\lim_{x \to 3} f(x) = \text{DNE}$$

Lastly, since we do not have the shaded dot towards the y-axis i.e. $f(3) = DNE$



REMOVABLE DISCONTINUITY

Removable discontinuities, also known as holes, occur when a function has a point where it's not defined, but the gap can be 'filled' to make the function continuous at that point.



Graphical Representation:
The graph will show an open circle or 'hole' at the discontinuity point.

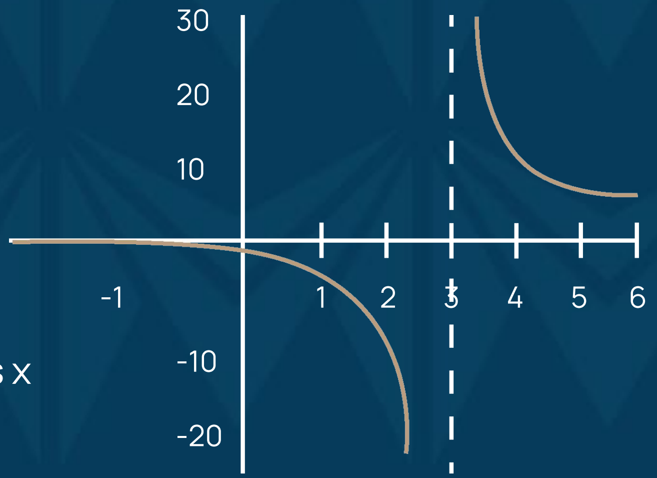
As x approaches the discontinuity point, the function approaches a particular value.

INFINITE DISCONTINUITY

Infinite discontinuities happen when a function approaches (positive) infinity or negative infinity as x approaches a specific point.

Graphical Representation:
The graph will show a vertical asymptote at the point of discontinuity.

The function rises or falls indefinitely as *x* approaches the point.



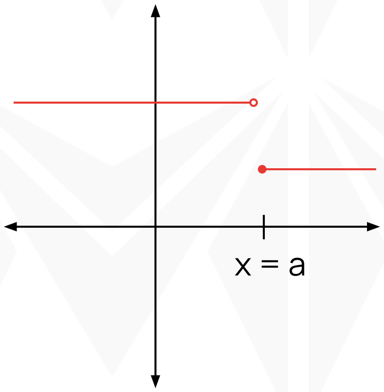
JUMP DISCONTINUITY

Jump discontinuities occur when the function has a sudden jump in value at a specific point.

Graphical Representation:
The graph will have a break or gap at the point of the discontinuity.

The function approaches different values from the left and right.

Jump Discontinuity



x = a

EXAMPLE 1

Function: $f(x) = \frac{x^2-1}{x-1}$

Limit: $\lim_{x\to 1} f(x)$

Graphical Analysis:
Factoring the function, we see it has a removable discontinuity at x = 1.

The graph will have a hole at x = 1, and we can determine the value the function approaches as x gets closer to 1.

QUESTION

From this function $f(x) = 2x + 1$, identify the slope (m) and y-intercept (c).

EVALUATING A BASIC LIMIT GRAPHICALLY

Given a function: $f(x) = 2x + 1$

The limit is written as: $\lim_{x \to 3} (2x + 1)$

This can be represented graphically as:

1. Plot the function $f(x) = 2x + 1$ on a coordinate plane. Draw a straight line with a slope of 2 (indicating that for every 1 unit to the right on the x-axis, the function rises by 2 units on the y-axis) and a y-intercept of 1 (where the line intersects the y-axis).

EVALUATING A BASIC LIMIT GRAPHICALLY

2. Locate the point x = 3 on the x-axis.

3. Observe how the function behaves as it approaches x = 3 on the graph. As x gets closer to 3, the function 2x + 1 rises smoothly along the line.

4. Since t**here are no abrupt breaks, holes, or jumps** in the graph as x approaches 3, it's evident that the function continues smoothly, and there are no disruptions in its behavior at
x = 3.

EVALUATING A BASIC LIMIT GRAPHICALLY

5. Therefore, the limit

$$\lim_{x \to 3} (2x + 1)$$

can be determined graphically by observing that the function approaches a single value as x gets closer to 3, which is $2(3) + 1 = 6 + 1 = 7$.

EVALUATING A ONE-SIDED LIMIT GRAPHICALLY

Given a function: $g(x) = |x|$

It's limit is given as: $\lim_{x \to 0^+} |x|$

GRAPHICAL REPRESENTATION

1. Plot the function $g(x) = |x|$ on a coordinate plane.

   Draw a V-shaped graph centered at the origin. It represents the absolute value function, which is symmetric around the y- axis.

2. Focus on the right side of the origin, where $x > 0$.

GRAPHICAL REPRESENTATION

3. Locate the point $x = 0$ on the x-axis.

4. Observe how the function behaves as it approaches $x = 0$ from the right side. As x gets closer to 0 from the right, the function approaches 0

5. Since the function smoothly approaches 0 as x approaches 0 from the right side, the one-sided limit $\lim_{x \to 0^+} |x|$ is 0.

NOTE

These examples demonstrate how graphical representation can provide an intuitive way to evaluate limits. By examining the behavior of the function on the graph as the input variable approaches a specific point, you can determine whether the limit exists and what its value is. In these cases, the graphical method aligns with the algebraic results.

SUMMARY

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- Limits define the behavior of a **function** at a particular point.

- Continuity implies a smooth, unbroken curve in the graph.

Graphical Analysis:

- Observe the graph near a point to determine if it's continuous, has holes, jumps, or asymptotes.

- Analyze how the function approaches the point from both sides.

FURTHER READING

Anton, H., Bivens, I., & Davis, S. (2016). *Calculus: Early Transcendentals*. John Wiley & Sons.

Courant, R., & John, F. (1988). *Introduction to Calculus and Analysis, Vol. I*. Springer.

Abbott, S. (2015). *Understanding Analysis*. Springer.

Thank You